Pedro Henrique França

Full Stack Developer

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Professional Summary

Full-stack developer with 3 years of experience building and running production web and mobile applications. I work with TypeScript, NestJS, React, React Native, Flutter, PHP, and Python in an educational technology ecosystem serving tens of thousands of monthly active users, and my scope has grown from features to the platform around them: infrastructure, cloud cost, application security, and AI features. I deliver most implementation through AI agents (Claude Code): I write the specifications, make the design decisions, and verify the result with tests, canary rollouts, and production checks.

Technical Skills

Languages: TypeScript, JavaScript (ES6+), PHP, Dart, Python, SQL, Rust

Backend & Data: NestJS, Prisma, TypeORM, MySQL/Aurora, CodeIgniter 3, Laravel 10 + Livewire, FastAPI, REST APIs, WebSocket (Socket.IO)

Frontend: React, Vite, MUI, Radix UI, TanStack Query, React Hook Form + Zod, SCSS, hls.js, Chart.js, Three.js

Mobile: React Native (Expo), Flutter, Apple StoreKit 2, Firebase, Drift/SQLite, offline-first architecture, WebView bridges, App Store Connect, Google Play Console

Cloud & Infrastructure: AWS (ECS, ECR, CloudFront, S3, Aurora, Cognito, SES, SQS, SSM, CloudWatch), Pulumi (IaC), Docker, GitHub Actions, Datadog, Clerk

Security: XSS, IDOR, and CSRF remediation, RBAC, signed cookies, Subresource Integrity, secrets hygiene

AI: Claude Code (spec-driven, multi-agent delivery), LLM integration (Chatbase, Gemini), versioned prompts

Testing & Quality: Jest, Vitest, Playwright, Cypress, React Testing Library, PHPUnit, flutter test, rule-to-test coverage gates, differential fuzzing, code review

Experience

Full Stack Developer

August 2024 – Present

TopInvest - PRIMO Group | Caxias do Sul, RS

PLATFORM AND ARCHITECTURE

- Designed and built a TypeScript exam SDK (sole author, ~300 commits) that replaced exam logic duplicated across five codebases. It runs in three production hosts (a legacy CodeIgniter app, a React LMS, and a Flutter app) behind a fail-closed canary rollout with deterministic cohorts and pinned-version rollback, served from a private CDN with signed cookies and Subresource Integrity.
- Researched and proposed the multi-tenancy architecture adopted in the LMS, implementing tenant isolation in the NestJS backend and hostname-based theming in React. The solution enabled a second white-label brand to operate on the same stack.
- Acted as a main contributor to the ecosystem's identity migration from AWS Cognito to an in-house SSO across TypeScript, PHP, Flutter, and React Native, including duplicate-account merging and real-time session invalidation.
- Became the main developer and release manager of a legacy CodeIgniter 3 application (half of all its commits in 2026), building a tree-based question engine, cross-device progress recovery, and B2B reports, and growing its test suite from 4 to 138 test files with characterization tests of legacy code.

RELIABILITY, COST, AND SECURITY

- Traced 96% of a CloudFront distribution's traffic (~US$420/month) to the mobile app re-downloading an uncompressed 27 MB catalog about 20 times per lesson, and fixed both server (compression and caching, 1.4 s to 25 ms) and app without breaking versions already in the stores. Moved Aurora to I/O-Optimized after computing the break-even point, saving ~US$215–260/month.
- Found the root cause of a daily outage in the staging ECS cluster (an account-wide patch scan exhausting memory on small hosts) and fixed it with a cgroup limit in Pulumi IaC, at no added cost and with production user data proven byte-identical.
- Remediated an independent security review of the exam SDK (stored XSS, IDOR, an API key shipped in HTML and app binaries) across four repositories. A 500-rule business-rules catalog I built for the LMS exposed an authentication-guard bypass and an IDOR leaking password hashes, both fixed in production.
- Found a TypeScript–PHP contract bug that corrupted restored answers while 847 unit tests stayed green, and designed a historical re-grade of 31,000 exam attempts validated by 24,000 differential fuzz cases against the production PHP grader, with zero divergences.

AI

- Shipped AI answer explanations inside exams, with quotas, paywall, and kill switch enforced on the server, and an LLM question-review pipeline: a Python/FastAPI prototype later moved into the admin panel as a queue with per-exam agents and prompts tied to official exam syllabi.
- Delivered most of my 2026 work through AI agents (Claude Code) with a spec, plan, subagent, and review workflow: 70+ versioned design specs, rules catalogs as executable specifications (373, 500, and 429 rules) with two-way coverage gates, and guardrails such as manual-only schema changes and read-only access to production data. From May to October 2026: 100 pull requests and 1,000+ commits.

MOBILE AND PRODUCT

- Built the Cursos TopInvest mobile app (React Native, Expo) as its sole developer, from proof of concept to release on the App Store and Google Play, with offline video in encrypted storage, push notifications, crash monitoring, and forced updates; resolved production video crashes through a planned SDK migration.
- Implemented payment and subscription integrations across mobile applications and backend services, including Apple StoreKit 2, PIX, boleto, QR code payments, transaction validation, and offline caching.
- Built NPS and lifecycle email subsystems with proof of sending, CRM and WhatsApp automations, and business-area exports over a 1.6M-row unindexed sales table using async workers and primary-key binary search.
- Developed an HP-12C financial calculator emulator in JavaScript, reproducing RPN and TVM calculations for an embeddable web application and Electron desktop build. Its lead-capture flow receives approximately 500 monthly users with an ~8% conversion rate.

Full Stack Intern

August 2023 – March 2024

FCODE | Caxias do Sul, RS

- Developed 6+ complete CRUD applications with CodeIgniter, jQuery, and MySQL, working from requirements gathering through production deployment.
- Modeled relational databases with 20 to 40 tables and built responsive interfaces with jQuery and Bootstrap.

Personal Projects

Cleanser — macOS storage cleanup tool

Rust, Rayon, SHA-256, Homebrew, CI/CD

Storage cleanup tool for macOS with GUI and TUI interfaces built on a shared Rust core. Designed the initial architecture and implemented dynamic file discovery, caching, risk classification, and duplicate detection using parallel SHA-256 hashing. Published as a Homebrew package with a CI/CD pipeline.

cleanser-claws.app

Cymatic — Real-time audiovisual synthesizer TypeScript, Web Audio API, WebGL2, GLSL, Vite

A real-time audiovisual synthesis system inspired by cymatics. Designed the application architecture and implemented configurable harmonic audio layers, radial interference rendering with WebGL, and synchronized audio-visual output through a shared AudioContext.

Live Demo

Education

Bachelor's Degree in Software Engineering March 2023 – August 2028 (in progress)

University of Caxias do Sul | Caxias do Sul, RS

Languages

Portuguese: Native

English: Fluent (technical documentation and written / spoken communication)